Fiscal Year:	FY 2015	**Task Last Updated:**	FY 01/15/2015
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Research Opportunities in Combustion Science		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	02/01/2010	**End Date:**	01/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:	ISS (for BASS) NOTE: End date is now 1/31/2016 per NSSC information (Ed., 1/14/2015) NOTE: End date is now 1/31/2015 per NSSC information (Ed., 3/12/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul (Universities Space Research Association) Olson, Sandra (NASA Glenn Research Center)		
Grant/Contract No.:	NNX10AD99A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e. 4 cm diameter) is chosen as a representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, shoulder and wake regions result in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with several imbedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples. In addition to James T'ien (PI), Dr. Paul Ferkul (NCSER) and Dr. Sandra Olson (NASA GRC) will participate as Co-Is in the project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	BASS-II experiment aboard the International Space Station was conducted successfully. By August, 2014, all the samples have been burnt. (a) Flame spread rate and extinction limits were obtained in purely forced flow in zero-gravity under various oxygen percentage for narrow Sibal fabric samples in a small wind tunnel inside the Microgravity Glovebox. Data are being analyzed and used to compare with model and to calibrate the model parameters. (b) Ignition test on 4-cm partial sphere was conducted in BASS II. Sustained ignition requires more than 1 minute application of the present igniter. Such a long time is necessary because of the long heat-up time of the bulk solid. This test will help us to device ignition strategies in the GEL experiment. (c) There is a plan to perform more tests using the partially burnt Sibal and PMMA sphere samples from the BASS I and II tests in ISS. (d) Two cast 4-cm spherical PMMA samples with 5 embedded thermocouples were customarily made. Preliminary tests in a flow tunnel with variable oxygen percentage showed the thermocouples behaves properly. Theoretical effort on the modeling of gas-phase radiation with spectral dependence is continuing. Requirement Definition Review (RDR) was conducted and completed in December 2014.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Johnston MC, T'ien JS, Muff DE, Zhao X, Olson SL, Ferkul PV. "Self induced buoyant blow off in upward flame spread on thin solid fuels." Fire Safety Journal. 2015 Jan;71:279-86. http://dx.doi.org/10.1016/j.firesaf.2014.11.007 , Jan-2015
Articles in Peer-reviewed Journals	Jomaas G, Torero JL, Eigenbrod C, Niehaus J, Olson SL, Ferkul PV, Legros G, Fernandez-Pello AC, Cowlard AJ, Rouvreau S, Smirnov N, Fujita O, T'ien JS, Ruff GA, Urban DL. "Fire safety in space – beyond flammability testing of small samples." Acta Astronautica. Available online 25 November 2014. In Press, Corrected Proof. http://dx.doi.org/10.1016/j.actaastro.2014.11.025 , Nov-2014
Articles in Peer-reviewed Journals	Zhao X, T'ien JS. "A three-dimensional transient model for flame growth and extinction in concurrent flows." Combustion and Flame. Available online 29 December 2014. In Press, Corrected Proof. http://dx.doi.org/10.1016/j.combustflame.2014.12.003 , Dec-2014